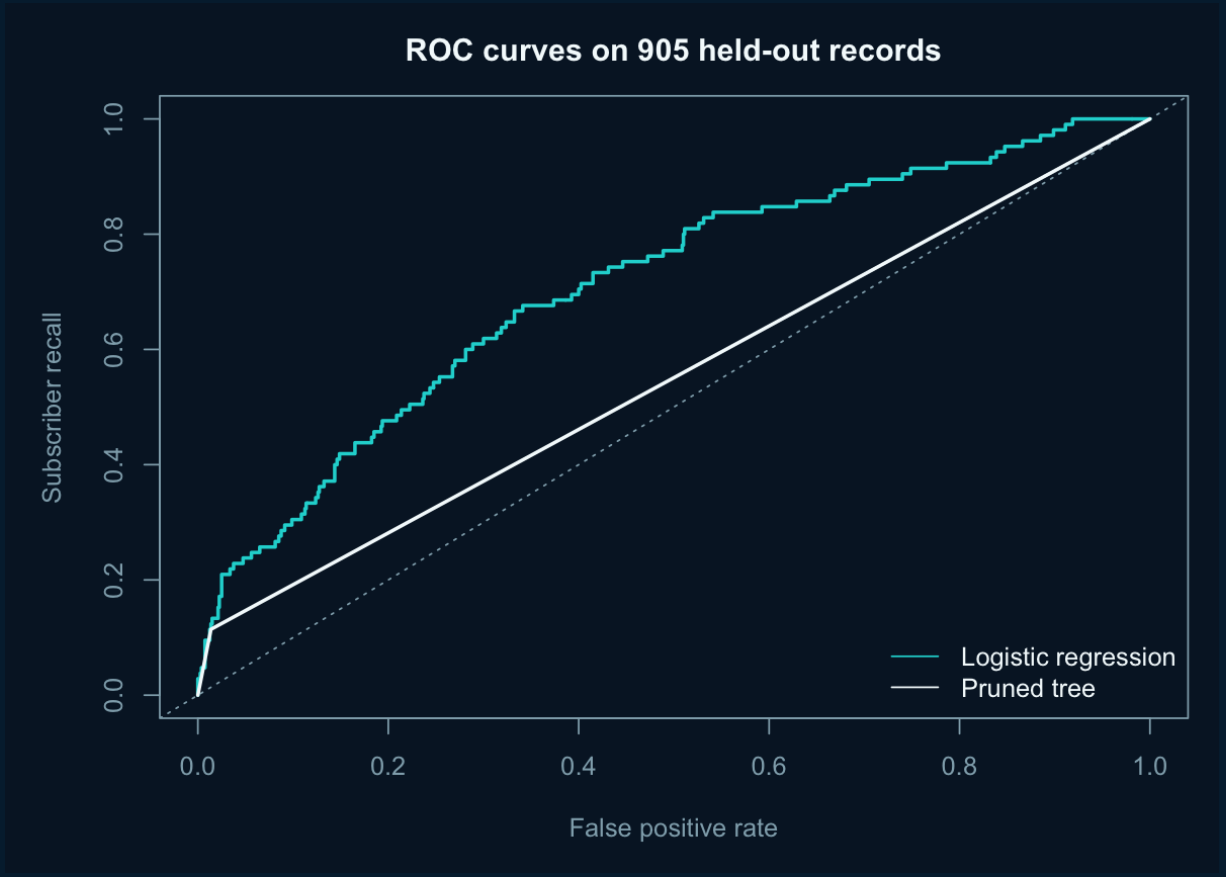
COMPLETED COURSEWORK / PORTFOLIO REVISION

Bank Marketing in R

Classification & model evaluation

Logistic regression | Pruned decision tree



Actual analysis output | Methods and limits inside

JAMES OOKO

A better question than overall accuracy

Can information available before a call identify term-deposit subscribers?

UCI Bank Marketing sample: 4,521 records. The analysis excludes call duration and current-campaign fields. Stratified splits: 2,712 training, 904 validation, and 905 test observations. The test set includes 105 subscribers.

The tree is pruned using cross-validation within training data. Probability thresholds maximize validation F1, then remain fixed for test evaluation.

Test performance	Accuracy	Precision	Recall	ROC AUC
Always predict no	88.4%	0.0%	0.0%	0.500
Logistic regression	84.3%	30.1%	26.7%	0.710
Pruned decision tree	88.5%	52.2%	11.4%	0.550

Precision and recall refer to subscribers. Logistic threshold: 0.17; tree threshold: 0.10. No test outcomes were used to choose thresholds.

What I learned from the comparison

The majority baseline is accurate but finds no subscribers.

Logistic regression finds more subscribers than the tree on this split, with more false positives. The tree keeps high overall accuracy but misses most subscribers. Neither result supports a strong production-performance claim.

What this demonstrates

Binary classification in R, prevention of decision-time leakage, stratified data splitting, validation-only threshold selection, and interpretation of class imbalance.

Limits

One random split is not chronological or external validation. F1 is a demonstration objective, not an estimated business cost. The data are historical, and any operational use would need additional data, stability, calibration, and appropriateness checks.

Source: Moro, S., Rita, P., and Cortez, P. (2014). Bank Marketing. UCI Machine Learning Repository. DOI: 10.24432/C5K306. Data license: CC BY 4.0. Uses the official bank.csv sample.

Code and full results: github.com/jamesooko/Final-Project-HOA732